

Exercise on Production Possibilities Model

Question 1

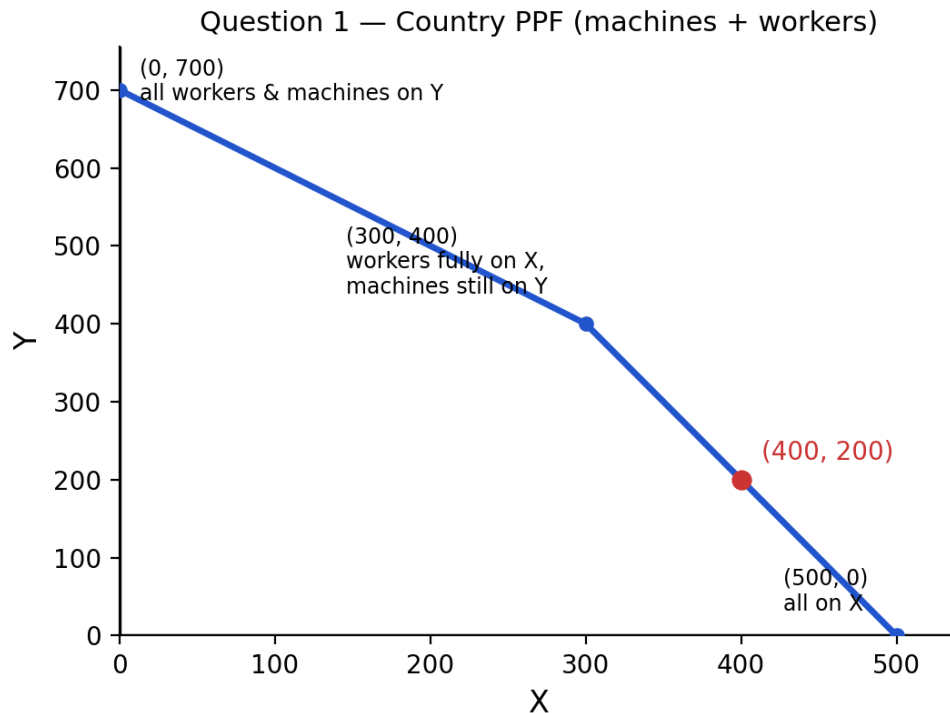
Suppose there is a country that can produce two goods: X and Y. The country has 100 machines and 50 workers. Each machine can produce 2 units of X or 4 units of Y. Each worker can produce 6 units of X or 6 units of Y.

- Who has a comparative advantage in producing X?
- Who has a comparative advantage in producing Y?
- Please draw a production possibilities frontier of this country.
- Suppose that this country needs to consume 400 units of X. In that case how many units of Y will be produced?

Answers:

	machines	workers
Production X per one unit	2	6
Production Y per one unit	4	6
MC in producing X in terms of Y	2	1
MC in producing Y in terms of X	0.5	1
Max X	$100 \cdot 2 = 200$	$50 \cdot 6 = 300$
Max Y	$100 \cdot 4 = 400$	$50 \cdot 6 = 300$

- Workers have a comparative advantage in producing X ($mc=1$ versus $mc=2$).
- Machines have a comparative advantage in producing Y ($mc=0.5$ versus $mc=1$).
- See answer c below.
- If a country needs to consume 400 units of X, then all the workers have to produce X (in total 300 units of X). Therefore 50 machines ($100/2$) will produce additional 100 units of X. The remaining 50 machines ($100-50$) will produce 200 units of Y ($50 \cdot 4$).



Question 2

Suppose there are two independent countries Zulu and Wulu. Those countries consume two goods, guns (X) and butter (Y).

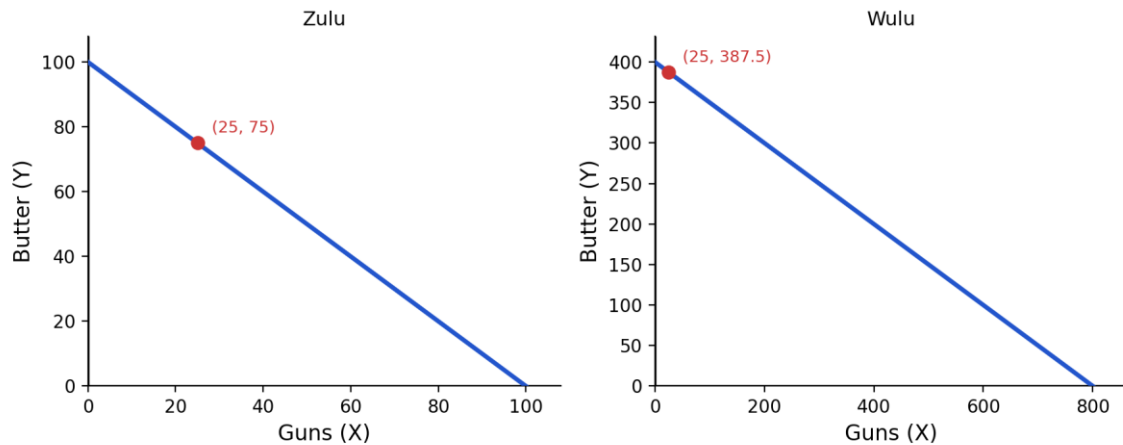
Assume that in Zulu there are 100 workers and each of them can produce 1 unit of guns or 1 unit of butter. Assume that in Wulu there are 400 workers and each of them can produce 2 units of guns or 1 unit of butter.

- Please draw a production possibilities frontier of each of these countries.
- It is known that each country needs to consume 25 guns for self-defence. Given that we are in a closed economy (i.e., no trades are allowed), how many units of butter will be consumed by each country?
- Now assume trades are allowed, such that 1 unit of butter is worth 1.5 units of guns. Please draw a production possibilities frontier of each of these countries.
- If each country still needs 25 units of guns, what will be the consumption of butter of each of these countries?
- Please explain whether trade is profitable or not for each of these countries.

Answers:

	Production of guns	Production of butter	MC guns	MC butter
Zulu	1	1	$1/1=1$	$1/1=1$
Wulu	2	1	$1/2$	2

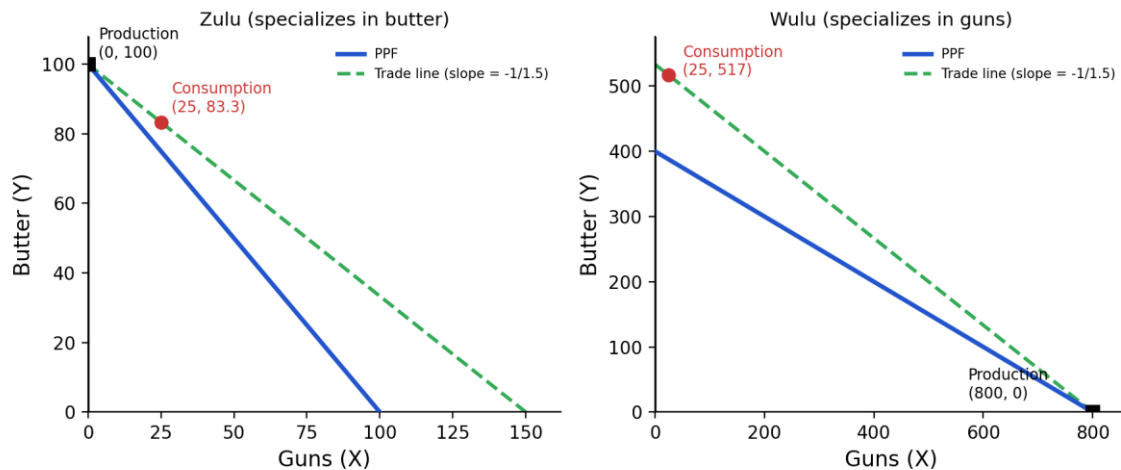
Question 2(a-b) — PPF without trade



b. To produce 25 units of guns Zulu needs 25 workers. The rest, 75 workers will produce 75 units of butter (75×1). In Wulu, 12.5 workers will produce guns and the rest 387.5 workers ($400 - 12.5$) will produce $387.5 \times 1 = 387.5$ units of butter.

c. For Wulu, it is worth to produce guns (cost of 0.5 units of butter per production of 1 unit of gun) and sell it to the world for the price of 0.67 units of butter ($1/1.5$). For Zulu, it is worth to produce butter (cost of 1 unit of gun per production of 1 unit of butter). Thus, Wulu will only produce 800 guns and trade it for butters (at most they can consume $800/1.5 = 533.3$ butter). Zulu will only produce 100 butter and trade it for guns (at most they can consume $100 \times 1.5 = 150$ guns).

Question 2(c-d) — PPF with trade (1 butter = 1.5 guns)



d. If each country still needs 25 units of guns, then Zulu will produce 100 units of butter and will sell 16.7 units of butter ($25/1.5$) for 25 units of guns. This means that they will consume 83.3 units of butter ($100-16.7$).

Wulu will produce 800 units of guns and keep to themselves 25 units of guns. The rest, 775 units of guns, they will sell for $775 \times 0.67 = 517$ units of butter.

e. We can see that with trade each country keeps on consuming the same number of guns, but larger amount of butter. This implies that trade is profitable for both countries.

Question 3

Explain the following sentence: After many years of a heavy military production on expense of technological development, many countries are not expected to have a sustainable economic growth.

Answer: Technological innovation increases productivity and therefore allows to expand the production possibilities frontier. Increase in productivity corresponds to productivity growth. Therefore, countries investing less in technological development will have a slower growth compared. However, there are some factors that negatively affect growth. Capital depreciation, population growth, increased wage inequality (i.e., capital more unequal), inequality between countries are all factors that could drastically slow down growth.

Useful links:

<https://ourworldindata.org/economic-growth>

<https://www.brookings.edu/blog/up-front/2020/02/25/technology-and-the-future-of-growth-challenges-of-change/>

https://datacommons.org/place/Earth?utm_medium=explore&mprop=amount&popt=EconomicActivity&cpv=activitySource%2CGrossDomesticProduction&hl=en